

Submission DSA1-FINAL-SUB05

Question DSA1-FINAL-Q01

alternate method (not the requested one):

rough work -> [arrow] [arrow]

I get path $4 \rightarrow 9 \rightarrow 6$; parent=6, but skipped the required derivation/justification.

Question DSA1-FINAL-Q02

alternate method (not the requested one):

rough work -> [arrow] [arrow]

I get hash set membership scan; expected $O(n)$ time, $O(n)$ space; prefix invariant, but skipped the required derivation/justification.

Question DSA1-FINAL-Q03

alternate method (not the requested one):

rough work -> [arrow] [arrow]

I get build $\Theta(n)$ with median recursion; search $\Theta(\log n)$ in balanced tree, but skipped the required derivation/justification.

Question DSA1-FINAL-Q04

alternate method (not the requested one):

rough work -> [arrow] [arrow]

I get A,B,C,D,E; BFS explores by distance layers and yields fewest edges, but skipped the required derivation/justification.

Question DSA1-FINAL-Q05

alternate method (not the requested one):

rough work -> [arrow] [arrow]

I get sorted prefix invariant; initialization, insertion maintenance, termination imply correctness, but skipped the required derivation/justification.

Question DSA1-FINAL-Q06

alternate method (not the requested one):

rough work -> [arrow] [arrow]

I get path $4 \rightarrow 9 \rightarrow 6$; parent=6, but skipped the required derivation/justification.

Question DSA1-FINAL-Q07

alternate method (not the requested one):

rough work -> [arrow] [arrow]

I get hash set membership scan; expected $O(n)$ time, $O(n)$ space; prefix invariant, but skipped the required derivation/justification.

Question DSA1-FINAL-Q08

alternate method (not the requested one):

rough work -> [arrow] [arrow]

I get build $\Theta(n)$ with median recursion; search $\Theta(\log n)$ in balanced tree, but skipped the required derivation/justification.

Question DSA1-FINAL-Q09

alternate method (not the requested one):

rough work -> [arrow] [arrow]

I get A,B,C,D,E; BFS explores by distance layers and yields fewest edges, but skipped the required derivation/justification.

Question DSA1-FINAL-Q10

alternate method (not the requested one):

rough work -> [arrow] [arrow]

I get sorted prefix invariant; initialization, insertion maintenance, termination imply correctness, but skipped the required derivation/justification.